Federal Board HSSC-I Examination
Model Question Paper Statistics
(Curriculum 2006)

ROLL NUMBER					
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

Version No.			
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Section - A (Marks 19)

Time Allowed: 25 minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. *Do not use lead pencil.*

Candidate Sign. _________ **Invigilator Sign.** ____________

Q1. Fill the relevant bubble against each question. Each part carries one mark

Sr no.	Question	A	B	C	D	A	B	C	D
i.	The branch of statistics that is concerned with procedures for obtaining valid conclusions is called:	Descriptive statistics	Inferential statistics	Theoretical statistics	Bio Statistics	○	○	○	○
ii.	Census reports published are:	Primary data	Secondary data	Confidential data	Fictitious data	○	○	○	○
iii.	A graph of a cumulative frequency distribution is called:	Histogram	Frequency polygon	Ogive	Frequency curve	○	○	○	○
iv.	The bases of classification are:	Two	Three	four	five	○	○	○	○
v.	Find the arithmetic mean of X if following deviations are given. $\sum(X-11)=6$, $\sum(X-30)=19$ and $\sum(X-17)=0$.	11	17	30	*19*	○	○	○	○
vi.	Geometric mean of numbers 2, 4 , 8 and 64 is:	4	8	16	64	○	○	○	○
vii.	The model letter(s) of the word “STATISTICS”:	S	S & I	S and T	T	○	○	○	○
viii.	The Harmonic Mean (H.M.) of the series 1, 2, 4 is:	5	7	7/5	12/7	○	○	○	○

ix.	The Sum of squared deviations of the values is least when deviations are taken from:	Athematic Mean	Median	Mode	Geometric mean	O O O O
x.	Variance is zero only if all observations are:	square	Square root	Different	Same	O O O O
xi.	The price relative is the percentage ratio of current year price and:	Current year price	Base year price	Current year quantity	Base year quantity	O O O O
xii.	An index number that measures the average relative change in cost of living is called:	Whole sale price index number	Consumer price index number	Sensitive price index number	Industrial index number	O O O O
xiii.	Which of the following indices has an upward bias?	Laspeyres Index number	Paasche index number	Fisher index number	Value index number	O O O O
xiv.	When regression line passes through origin then intercept 'a' is:	One	zero	negative	positive	O O O O
xv.	The Independent variable is also called:	Explanatory variable	regressor	predictor	All of these.	O O O O
xvi.	When two variables move in same direction, the correlation will be:	zero	positive	negative	neutral	O O O O
xvii.	The value of correlation coefficient (r) lies between:	$-\infty$ to ∞	-1 to 1	0 to 1	-1 to 0	O O O O
xviii.	In summers, the sale of ice-cream increases is an example of:	Secular variation	Cyclical variations	Seasonal variations	Random variation	O O O O
xix.	The main components of time series are:	two	three	four	five	O O O O

Federal Board HSSC-I Examination Model Question Paper Mathematics (Curriculum 2006)

Time allowed: 2.35 hours Total Marks: 66

Note: Answer all parts from Section 'B' and all questions from Section 'C' on the **E-sheet**. Write your answers on the allotted/given spaces.

SECTION – B (Marks 42) $(14 \times 3 = 42)$

Q.2

i. Following are the number of mistakes made by 25 typists in a typing test:

4, 8, 0, 5, 8, 4, 3, 5, 7, 7, 8, 8, 3, 2, 7, 1, 1, 0, 2, 4, 6,4,6,3,5. Make a discrete frequency distribution. (M: 3)

OR Distinguish between histogram and historigram. (M: 1.5+1.5)

ii. Make class boundaries, class mark and cumulative frequency for the following frequency distribution. (M: 1+1+1)

Classes	Frequency	Classes	Frequency
3-3.9	5	6-6.9	10
4-4.9	7	7-7.9	5
5-5.9	8	8-8.9	3

OR If $\sum_{i=1}^{4} X$=10, $\sum_{i=1}^{4} Y$ =40, $\sum_{i=1}^{4} XY$=100. Find the values of $\sum_{i=1}^{4}(2X+5)(Y-4)$) (M: 3)

iii. Define discrete and continuous variable with examples. (M: 1.5+1.5)

OR Write down any three qualities of a good average (M: 3)

iv. The average marks obtained by three sections of 1^{st} year class are given below. Find the combined mean of 1^{st} year class. (M: 3)

Sections	No. of students	Means
A	35	60
B	30	50
C	20	70

OR The reciprocal of the values of the variable 'X' are given below. Compute mean of X.
$\frac{1}{X}$ =0.0833, 0.0526, 0.0667, 0.0588, 0.05, 0.0370, 0.0455. (M: 1+2)

v. The following values have been obtained from a frequency distribution of a variable x.
X=62+5U, $\sum f$=120, $\sum fU$ = 140, $\sum fU^2$=598.
Calculate arithmetic mean, variance and standard deviation (M: 1+1+1)

OR The following data are obtained from past series of innings of two batsmen A and B. (M: 1+2)

	Batsman A	Batsman B
Average inning Score	75	60
Variance of score	64	25

(a)Which batsman is better run getter
(b) Who is more consistent player

vi. In moderately skewed distribution, mean is 25 and mode is 28. Find the approximate value of median (M: 3)

OR Compute Mean deviation from median and Mean coefficient of dispersion from the following.
8, 4, 6 , 5 , 9 ,10 , 2 , 13 ,15 (M: 2+1)

vii. Calculate Index number (a) taking 2010 (b) average of first 03 years as base. (M: 1.5+1.5)

YEAR	2010	2011	2012	2013	2014	2015	2016
PRICE	20	18	25	28	30	35	42

OR For a frequency distribution of 'X', it is given that Q_1 =20, Q_2= 40 and Q_3=50. Find the Bowley's coefficient of skewness (M: 1+2)

viii. If V(X)=4 and V(Y)=5 ,then find the
(a) Var(2X-3Y)
(b) $Var(\frac{X}{2}+5Y)$ (M: 1.5+1.5)

OR If fisher index number = 156 and laspeyres index number =160 then find the value of Paasche index number. (M: 3)

ix. Compute consumer price index number from the following. (3)

Commodity	A	B	C	D	E
Weight(W)	6	5	4	1	2
Price relative(I)	61.54	146.67	74.3	76.22	90.14

OR Differentiate between simple and composite index numbers (1.5+1.5)

x. If the regression lines Y on X and X on Y are 2X-3Y=0 and 4y-5X=0 then find b_{yx} and b_{xy}. (1.5+1.5)

OR Describe the properties of correlation coefficient(r). (3)

xi. **If b_{yx} = -0.95 and b_{xy} = - 0.8. Find the value of correlation coefficient(r).** (3)

OR **If n=10 , $\sum Dx$= -87 , $\sum Dy$= 43 , $\sum D^2x$= 949 , $\sum D^2y$=543 , $\sum DxDy$=385 .Find Coefficient of correlation (r)** (3)

xii. Calculate 3-years simple moving average from the following. (3)

Years	2000	2001	2002	2003	2004	2005	2006	2007
Values	20	22	26	25	28	24	30	34

OR Given $\bar{X} = 50$ *and* $\bar{Y} = 60$, S_{yx}=36, S_x=4 and S_y=9. Find regression line X on Y. (3)

xiii. Describe seasonal variation with examples. (2+1)

OR Differentiate between regression and correlation. (1.5+1.5)

xiv If $\sum X=0$, $\sum Y=245$, $\sum XY=66$, $\sum X^2=28$ and n=7 .Fit a linear trend. (3)

OR Define 'signal' and 'noise' (1.5+1.5)

SECTION – C (Marks 24)(4×6=24)

Note: Attempt all questions.

Q3 Calculate Arithmetic Mean, Median and Mode of the following frequency distribution. (Marks 6)

Classes	f	Classes	f
9.3-9.7	2	11.3-11.7	14
9.8-10.2	5	11.8-12.2	6
10.3-10.7	12	12.3-12.7	3
10.8-11.2	17	12.8-13.2	1

OR Calculate chain indices for the following taking Median as an average (Marks 6)

Year	Wheat	Rice	sugar	Tea
1980	60	40	20	200
1981	80	45	30	220
1982	100	48	35	250
1983	120	50	50	300

Q4 Compute Laspeyres's, Paasche's and Fisher's index numbers for 2003 taking 2000 as base. (Marks 6)

Commodity	2000		2003	
	Price	Quantity	Price	Quantity
A	40	10	50	12
B	37	15	40	18
C	27	20	30	25
D	30	12	34	16

OR Find variance, standard deviation and coefficient of variation of the following. (Marks 6)

Classes	f
0 - 10	6
10- 20	8
20-30	10
30-40	7
40-50	5
50-60	4

Q5 Find regression line Y on X and estimate Y when X=10 from the following. (Marks 6)

X	17	13	15	16	6	11	14	9	7
Y	36	46	35	24	12	18	27	22	8

OR Find the trend values by the method of semi-average from the following data. (Marks 6)

Year	1990	1991	1992	1993	1994	1995	1996	1997
values	97	85	100	90	105	83	112	120

<table>
<tr>
<td>Q6</td>
<td>Compute 4 year centered moving average for the following time series.

<table>
<tr><th>year</th><th>Production in million (Kg)</th><th>year</th><th>Production in million (Kg)</th></tr>
<tr><td>1993
1994
1995
1996</td><td>331
344
349
332</td><td>1997
1998
199
2000</td><td>364
395
400
410</td></tr>
</table>
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<td>6</td>
<td>O
R</td>
<td>Determine the value of correlation (r) between number of absentees (X) and final marks (Y) and interpret the result.
<table>
<tr><td>X</td><td>6</td><td>2</td><td>15</td><td>9</td><td>12</td><td>5</td><td>8</td><td>10</td></tr>
<tr><td>Y</td><td>82</td><td>96</td><td>43</td><td>74</td><td>58</td><td>90</td><td>78</td><td>60</td></tr>
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Table of Specification

Model Question Paper STATISTICS – Grade XI (HSSC-I)

(Curriculum 2006)

Topics	1 INTRODUCTION TO STATISTICS	2 PRESENTATION OF DATA	3 MEASURES OF LOCATION	4 MEASURES OF DISPERSSION	5 INDEX NUMBER	6 REGRESSION & CORRELATION	7 TIME SERIES	Total marks of each assessment objectives	Percentage of Cognitive Level
Knowledge									17%
Comprehension (Understanding)									44%
Application									39%

Key:

- 1, 2, 3 etc. stands for question numbers
- i, ii, iii etc. stands for part of question numbers
- (1), (2), (3) etc. stands for marks of question papers
- Question Number (part/ first choice) marks example: Q2 (i / f) 4
- Question Number (part/ second choice) marks example: Q2 (i / s) 4

Note:

1 This TOS does not reflect policy, but it is particular to this model question paper.

2 Proportionate / equitable representation of the content areas may be ensured.

3 The percentage of cognitive level is 30%, 50%, and 20% for knowledge, understanding, and application, respectively with **± 5%** variation.

4 While selecting alternative questions for SRQs and ERQs, it must be kept in mind that:

- Difficulty levels of both questions should also be same
- SLOs of both the alternative questions must be different